

OPEN SCAFFOLD LABS · VERTICAL APPLICATION BLUEPRINT

TugOS is App 24 in the OSL Ecosystem

The Open Scaffold Orchestrator Model isn't just for ad agencies and fire departments. Every architectural primitive — the three-surface model, the MCP layer, the Catalog of Authority, the genetic template — maps directly onto tugboat operations with zero framework changes.

React 19 + Vite Express.js + Vercel Supabase + PostgreSQL 5 MCP Servers Signal K · OpenCPN
· AIS Table Prefix: tug_

01 · THE GENETIC TEMPLATE — DEPLOY IN UNDER 5 MINUTES

FRAMEWORK AS A SERVICE (FAAS) — SHARED ACROSS ALL 23 OSL APPS

@openscaffold/core · @openscaffold/integrations · scaffold.config.js

One genetic template. Run `deploy-new-app.sh tug_` and TugOS inherits: StandardHeader, Scaffold Beacon AI widget, role-based sidebar, JWT auth, theme system, CRUD generation standards, Supabase connection, and all integration adapters (pdf-lib, PptxGenJS, React Email, DocuSeal, LlamaIndex).

TugOS is App 24. Table prefix: `tug_`. Shares one Supabase PostgreSQL instance with all 23 existing apps — zero new infrastructure provisioning required.



DOMAIN LAYER

TugOS Domain UI

Dispatch · Crew · Fuel · Maintenance · Billing

The tug-specific screens and workflows — what makes TugOS TugOS. Built using the OSL component library but configured for maritime operations.

Vessel dispatch board, crew hitch scheduler, per-voyage fuel log, USCG compliance tracker, invoice

generator EXPERTISE LAYER

EXPERTISE LAYER

Catalog of Authority

EaaS — Maritime Authorities

No tug domain expert hired. Instead, validated MCP connections to the authorities those experts would consult.

INTELLIGENCE LAYER

Paperclip + Claude API

AI Orchestration — OSL Native

OSL's existing AI orchestration framework, already running in production. TugOS plugs in without rebuilding.

Fuel anomaly detection · Maintenance prediction · Crew cert risk scoring · Auto-invoice from job log ·
Dispatch optimization

02 · THE THREE-SURFACE ARCHITECTURE — EVERY OSL APP, INCLUDING TUGOS

01

Surface 1: The UI

Human-Facing · React 19 + Vite + Tailwind

The [React](#) 19 browser app. Role-filtered sidebar — Dispatcher sees the job board, Captain sees their vessel log, Fleet Manager sees the P&L dashboard. The Scaffold [Beacon](#) floats on every page as the TugOS AI assistant.

TugOS Roles: Dispatcher · Captain · Crew · Port Captain · Fleet Admin · Billing Clerk · Client (read-only)

02

Surface 2: The REST API

Programmatic · Express.js + Vercel Serverless

[Express.js](#) deployed as [Vercel](#) serverless functions. Every UI action is an API call. Also serves: AIS webhooks from MarineTraffic, Signal K vessel data pushes, QuickBooks sync callbacks, port authority job intake feeds.

TugOS API Consumers: TugOS web app · React Native captain app · Port authority systems · Signal K IoT bridge · QuickBooks / Xero sync

03

Surface 3: The MCP Layer

AI-Native · Bidirectional Authority + Agent Access

As [MCP](#) Client: TugOS consumes maritime authority expertise from the Catalog of Authority. As MCP Server: TugOS exposes its fleet data to AI agents. Same pattern as Open Agency — three views of one application.

MCP Inbound: USCG regs · AIS feed · Weather · Signal K **MCP Outbound:** Fleet status · Crew records ·
Job logs · Fuel data



THE BRAIN · 20+ TOOLS

tug-orchestration

OSL equivalent: open-orchestration — agent hub, CRM, Catalog of Authority

TugOS tools: list_vessels, get_vessel_status, list_active_jobs, get_crew_roster, trigger_dispatch, approve_fuel_alert, get_fleet_health, search_authority_catalog (USCG/STCW), get_job_briefing, generate_invoice



THE SENSES · 15+ TOOLS

tug-data-connectors

OSL equivalent: mcp-adtech-connectors — normalizes data across 4 ad platforms

TugOS tools: get_ais_position, get_signal_k_data (fuel flow, engine hours, GPS), get_weather_noaa, get_port_schedule, get_marinetraffic_eta, get_fuel_price, get_tidal_data, sync_quickbooks



THE VOICE · 12+ TOOLS

tug-notifications

OSL equivalent: mcp-notifications — Slack, email, iMessage dispatch

TugOS tools: send_captain_push (mobile), send_dispatch_slack, send_cert_expiry_alert, send_fuel_anomaly_alert, send_uscg_reminder, send_client_job_complete, send_maintenance_urgent, send_invoice_ready



THE MEMORY · 15+ TOOLS

tug-scaffold-core

OSL equivalent: mcp-scaffold-core — Catalog of Authority, ecosystem knowledge

TugOS tools: search_uscg_regulations, get_stcw_requirements, get_abs_survey_standards, **TugOS tools:** search_uscg_regulations, get_stcw_requirements, get_abs_survey_standards, get_imo_compliance, list_vessels (all 23 app pattern), get_architecture_pattern, search_playbook, get_maritime_authority



THE HANDSHAKE · 8 TOOLS

tug-client-portal

OSL equivalent: mcp-client-portal — client-scoped read-only, authenticated per client

TugOS tools (scoped to client): my_jobs, my_job_history, my_invoices, my_invoice_status, my_vessels_assisted, my_compliance_certs, my_account_health, download_job_report



PERSISTENT MEMORY · 6TH LAYER

NotebookLM — Maritime Knowledge

OSL equivalent: NotebookLM — survives context compression, session-to-session memory

TugOS notebooks: USCG Inspection Playbook (regulatory history) · Fleet Maintenance

04 · SIX AI AGENTS — MARITIME EQUIVALENTS OF OPEN AGENCY'S AGENT SUITE

Fuel Agent AUTO-LIMITED OSL: Yield Agent — budget shifts < \$5K, Databricks_Spark_Sim

Monitors per-vessel fuel consumption vs baseline. Flags anomalies > 8%. Can approve fuel efficiency adjustments under \$2K threshold. High-cost alerts route to fleet manager via Interrupt Protocol.

Compliance Agent SUPERVISED OSL: Attribution Agent — supervised, statistical analysis

Scans all crew certification expiry dates daily. Generates USCG manning compliance reports. Flags vessels at risk of manning violations. All recommendations require Port Captain review before notification sent to crew.

Dispatch Agent SUPERVISED OSL: Audience Agent — identity/segmentation, supervised

Recommends optimal vessel assignment for incoming assist jobs based on proximity (AIS), crew availability, bollard pull requirements, and fuel cost. Dispatcher approves before push notification sent to captain.

Fleet Health Agent AUTONOMOUS OSL: Health Agent — client monitoring, autonomous

Runs hourly. Scores each vessel on maintenance status, fuel efficiency trend, crew compliance, and utilization rate. Pushes alerts to fleet manager Slack when any vessel drops below 70 health score.

Daily Briefing Agent AUTONOMOUS OSL: Digest Agent — daily briefing compilation, autonomous

Runs at 6am daily. Compiles: today's scheduled jobs, crew going on/off hitch, certifications expiring this week, maintenance due, fuel cost vs budget, open invoices. Delivered as email + Slack to fleet manager.

Invoice Agent AUTONOMOUS OSL: Priority Agent — task routing, autonomous

Monitors completed jobs. Auto-generates draft invoices from job logs within 1 hour of job completion. Pushes draft to billing clerk for 1-click approval. Captures billable events that would otherwise be manually compiled days later.



The Interrupt Protocol — Unchanged from Open Agency

High-stakes TugOS actions hit an `interrupt_before` node in the LangGraph workflow exactly as Open Agency does. Port Captain reviews in the TugOS War Room with full execution trace before action fires. Triggers: fuel procurement > \$5K · new crew certification waiver · client billing dispute resolution · USCG deviation reporting · any PII crew record change. LangGraph + MLflow governance is identical — OSL brings this to TugOS at zero additional framework cost.

05 · SEVEN SCHEDULED AUTOMATIONS — CRON-BASED VIA CLAUDE SCHEDULED TASKS

TASK NAME	SCHEDULE	WHAT IT DOES	CONNECTORS
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daily-fleet-briefing	6am daily	Compiles jobs, crew changes, cert expirations, fuel budget, open invoices → email + Slack to fleet manager	Orchestration → Notifications
vessel-health-monitor	Hourly	Scores fleet health, sends Slack alert when vessel drops below 70, urgent push for critical maintenance	Orchestration → Notifications → Captain App
cert-expiry-sweep	7am daily	Scans all STCW / MMC / drug test / medical cert dates across crew roster. Queues alerts at 90/30/7 day windows	Orchestration → Notifications → Email
invoice-generation	Every 60 min	Finds completed jobs with no invoice. Generates draft from job log. Pushes to billing queue for 1-click approval	Orchestration → Notifications → QuickBooks
authority-catalog-sync	6am daily	Syncs PostgreSQL maritime authority catalog → Notion database. Team can browse USCG/STCW rules without Claude	Orchestration → Notion MCP
fuel-anomaly-check	After each voyage	Compares voyage fuel consumption to vessel baseline. Flags >8% deviation to Fleet Agent for review	Signal K → Orchestration → Notifications

ais-position-sync	Every 5 min	Pulls live AIS positions for all fleet vessels. Updates dispatch map. Calculates ETA for active jobs	MarineTraffic API → Data Connectors → UI WebSocket
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06 · THE CATALOG OF AUTHORITY — TUGOS MARITIME AUTHORITIES



Regulatory Authorities

USCG · IMO · ABS · STCW

› USCG 33 CFR

› STCW

Convention

— Navigation and Navigable Waters; crew manning requirements, vessel inspection intervals

— International Standards of Training, Certification

› IMO MARPOL — fuel oil record book requirements, emission compliance, CII rating methodology › 46

CFR Part 15 — Manning of vessels; required licenses and endorsements by vessel type and route

and Watchkeeping; competency requirements per rank

› ABS Rules for Building and Classing Steel Vessels

— survey intervals, maintenance standards, classification society requirements



Data Authorities

Signal K · AIS · NOAA · OpenCPN

- › **Signal K (Apache 2.0)**

— Open marine data standard; real-time vessel telemetry from fuel flow meters, GPS, engine sensors
- › **MarineTraffic AIS Toolbox (Open Source)**

— Live vessel positions, port calls, ETA computation from AIS transponder data
- › **NOAA Weather & Tides API**

— Official U.S. weather and tidal data for operational planning and safety ›
- OpenCPN (GPLv2)**

— Chart data and navigation primitives for the dispatch map overlay
- › **OpenShipping.org API Standards**



Market Rates

— Open shipping logistics API protocol for port authority job intake integration

Financial Authorities [QuickBooks](#) · [Stripe](#) ·

- › **QuickBooks API**

— Two-way sync for existing customers; invoices, payments, chart of accounts ›
- Stripe**

— [Client portal](#) payments; card on file, ACH for large invoices
- › **Bunker fuel price feeds**

— Real-time marine diesel pricing by port from [Ship & Bunker API](#) › **AWO Rate**

— American Waterways Operators industry billing rate data for assist tug market
- › **AWO Rate Benchmarks**

— American Waterways Operators industry billing rate data for assist tug market pricing



OSL Native Integrations
Reused from [@openscaffold/integrations](#)

- › **pdf-lib (MIT)**

— USCG inspection reports, maintenance work orders, invoice PDFs — zero new code ›
- PptxGenJS (MIT)**

— Fleet performance reports as PowerPoint for client presentations
- › **React Email**

— Transactional email templates: job confirmations, cert alerts, invoice delivery ›
- DocuSeal (AGPL-3.0)**

— Digital signature capture for client job sign-off, isolated by adapter pattern ›
- LlamaIndex**

— RAG over USCG regulations and maintenance history for the Beacon AI assistant

07 · SHARED TECHNOLOGY STACK — WHAT TUGOS INHERITS VS WHAT'S NEW

LAYER	TECHNOLOGY	TUGOS USAGE
Frontend REUSE	React 19 + Vite 5 + Tailwind CSS	Dispatch board, crew scheduler, fuel dashboard, billing module — built on OSL component library. Dark maritime theme (navy/tide palette).

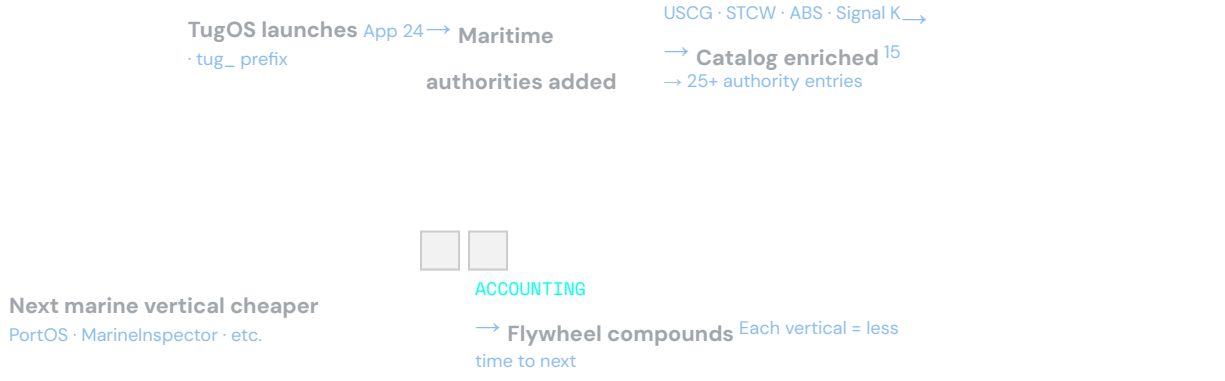
Mobile App NEW	React Native (Expo) – iOS first	Captain and crew app. Shared business logic with web. Offline-first architecture for vessel connectivity gaps. Job acceptance, fuel log, photo capture, digital signature.
Backend REUSE	Express.js 4 + Vercel Serverless	Same pattern as all 23 OSL apps. Direct SQL queries, no ORM. tug_ table prefix on shared Supabase instance.
Database REUSE	PostgreSQL on Supabase	One shared instance. New tables: tug_vessels, tug_jobs, tug_crew, tug_fuel_logs, tug_maintenance, tug_invoices, tug_certs, tug_clients.
Auth REUSE	JWT + bcrypt · 15-min tokens	Roles: fleet_admin, port_captain, dispatcher, captain, crew, billing, client. requireRole middleware unchanged from OSL pattern.
AI Service REUSE	callAI utility + Paperclip	OSL's existing dual-model setup (primary + Claude fallback) powers TugOS Beacon. Paperclip orchestrates the 6 maritime agents.
IoT Bridge NEW	Signal K Server (Apache 2.0)	Open marine data standard. Connects fuel flow meters, GPS units, and engine monitors on each vessel to TugOS via WebSocket. Critical for real-time fuel tracking.
	MarineTraffic	Real-time vessel positions overlaid on dispatch map. ETA

AIS Tracking NEW	API + Open AIS Toolbox	calculation for active assist jobs. Open-source AIS decoder for self-hosted fallback.
Agent Framework REUSE	LangGraph + MLflow	Same governance architecture as Open Agency. Interrupt Protocol for high-stakes maritime decisions. All agent runs logged to tug_agent_runs.
Deployment REUSE	Vercel + Supabase + GitHub	Existing OSL CI/CD pipeline. Auto-deploy on push. No new infrastructure. Fly.io for compute-heavy Signal K bridge workers if needed.

The Catalog of Authority Flywheel

Each new OSL vertical enriches the catalog. TugOS adds maritime authorities that benefit every future marine vertical.





09 · BUILD VS REUSE — THE HONEST

✓ INHERITED FROM OSL GENETIC TEMPLATE + NET NEW FOR TUGOS

- › StandardHeader, Scaffold Beacon AI widget
- › Role-based sidebar + requireRole middleware
- › JWT auth + bcrypt + session management
- › Supabase connection + Row-Level Security
- › All @openscaffold/integrations adapters › pdf-lib, PptxGenJS, React Email, DocuSeal
- › Vercel deployment pipeline + GitHub CI/CD
- › LangGraph + MLflow agent governance
- › Paperclip AI orchestration framework › MCP server SDK + 5-server pattern
- › Interrupt Protocol (LangGraph interrupt_before)
- › callAI utility (dual-model: GPT-4o + Claude)
- › Theme system + Sacred Cube branding
- › CRUD generation standards

- › Signal K IoT bridge (fuel sensors, GPS, engine)
- › MarineTraffic AIS integration + dispatch map
- › React Native captain/crew mobile app
- › Maritime Catalog of Authority entries (USCG, STCW, ABS, IMO)
- › tug_ database schema (8 new tables)
- › Dispatch board UI (drag-and-drop job assignment)
- › Hitch scheduler (crew rotation logic) › USCG compliance tracker module › Per-voyage fuel log UI + anomaly logic
- › Tug-specific billing engine (hourly / bollard pull / charter)
- › 6 maritime AI agents (Fuel, Compliance, Dispatch, Health, Briefing, Invoice)
- › QuickBooks two-way sync adapter › NOAA weather + tidal data connector

